Bank Account: Hardhat Commands and Code

Nov 4, 2026 · @Shreyash

Step 1: Install Node.js and npm

```
cd ~
sudo apt update
sudo apt install -y nodejs npm
hash -r
node -v
npm -v
```

Step 2: Create the project and install the tools

```
cd ~
mkdir Bank_Account
cd Bank_Account
mkdir contracts scripts
npm init -y
npm install hardhat@2 @nomicfoundation/hardhat-ethers@3 ethers@6 solc@0.8.24
ls
```

If Hardhat is already installed in Bank_contract

Use this in place of the `npm install` line. This example copies from `Bank_contract`. If the first command says `No such file or directory`, use the `npm install` line instead.

```
ls ~/Bank_contract/node_modules/hardhat
cp -r ~/Bank_contract/node_modules ~/Bank_Account/
ls
```

Step 3: Create hardhat.config.js

```
code hardhat.config.js
```

```
require("@nomicfoundation/hardhat-ethers");

module.exports = {
  solidity: "0.8.24"
};
```

Step 4: Create contracts/BankAccount.sol

```
code contracts/BankAccount.sol
```

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.24;

contract BankAccount {
    mapping(address => uint256) private balances;

    function deposit() public payable {
        require(msg.value > 0, "Send some Ether");
        balances[msg.sender] += msg.value;
    }

    function withdraw(uint256 amount) public {
        require(amount > 0, "Amount must be greater than zero");
        require(balances[msg.sender] >= amount, "Insufficient balance");

        balances[msg.sender] -= amount;
        payable(msg.sender).transfer(amount);
    }

    function showBalance() public view returns (uint256) {
        return balances[msg.sender];
    }
}
```

Step 5: Compile

```
npx hardhat compile
```

```
Compiled 1 Solidity file successfully (evm target: paris).
```

Step 6: Create scripts/run.js

```
code scripts/run.js
```

```
const { ethers } = require("hardhat");

async function main() {
  const bank = await ethers.deployContract("BankAccount");

  await (await bank.deposit({ value: 500 })).wait();
  console.log("After deposit:", await bank.showBalance());

  await (await bank.withdraw(200)).wait();
  console.log("After withdraw:", await bank.showBalance());
}

main();
```

Step 7: Execute

```
npx hardhat run scripts/run.js
```

```
After deposit: 500n
After withdraw: 300n
```